

An Open-Source PCB Test Vehicle for Electrical Characterization of Solder-Bonded Micro-LEDs: Design, Schematic, and Measurement Workflow

Course report — ET4277 + ET4391 — Part 1

Daniel Tyukov

Department of Microelectronics, ECTM + ITEC

Delft University of Technology

Delft, the Netherlands

Student no. 5714699 | daniel.tyukov@alphait.us



Abstract—We present the design and intended measurement workflow of a 93 mm × 93 mm two-layer FR-4 test vehicle for electrical characterization of solder-paste-bonded micro-LEDs. The board integrates 26 Würth WL-SFCC 0404 RGB LED footprints (8 individually addressable + 6+12 in two series chains), four 0402 NTC thermistors for forward-voltage thermal sensing parameter (V_F -TSP) thermometry, a $100\ \Omega \pm 0.1\%$ precision resistor with companion OPEN/SHORT/LOAD calibration pads for electrochemical impedance spectroscopy (EIS), three transmission-line-method (TLM) ladders and four van der Pauw (VDP) cloverleaves for contact-resistivity and sheet-resistance extraction, and a 6×6 design-of-experiments (DoE) bond-pad array. Two 32-pin breadboard-compatible headers expose every active signal. The board is fab-ready for the Aisler standard pool with ENIG finish (0 DRC violations); LEDs are intentionally not populated by the fab so the research subject (the bond joint itself) can be formed under controlled cleanroom conditions. We describe the schematic, the four core measurement methods with their governing equations, the tooling required, and the solder-mixture experiment plan that motivates the design.

Index Terms—micro-LED, die bonding, solder paste, V_F -TSP, electrochemical impedance spectroscopy, transmission line method, van der Pauw, daisy chain, electrical characterization, PCB test vehicle

I. INTRODUCTION

Micro-LED display and lighting modules require sub-millimeter dies to be electrically and mechanically joined to a substrate at high yield, low voltage drop, and low thermal resistance. Reflow-soldering with conventional SAC305 paste is the cheapest route, but the joint quality depends strongly on paste rheology, stencil aperture, reflow profile, and substrate finish [1], [2]. Alternative chemistries (Au-Sn eutectic [3], Cu-Cu [4], anisotropic solder films [5], polymer-reinforced pastes [1]) trade off process temperature, contact resistivity, and impact reliability.

This work extends the v1 ECTC-2025 board of Abdelwahab et al. [6] into a v2 platform that supports four orthogonal electrical-characterization methods on a single PCB so that one bonding trial yields self-consistent bond, contact, sheet, and thermal data. We target the Aisler standard pool to keep the artifact reproducible and cheap (~40 per assembled board).

The remainder of the paper covers the board architecture (Section II), the schematic blocks and net plan (Section III), the four characterization methods with their governing equations and the specific footprints they exercise (Section IV), the tooling and assembly workflow (Section V), and the solder-mixture experiment that motivated the design (Section VI).

II. BOARD ARCHITECTURE (v4.0.7)

The board (Fig. 1) is laid out in five horizontal zones from top to bottom: (a) a title block carrying project, designer, and course metadata; (b) the 32-pin north header (Würth WR-PHD 61304011121, 1×40 single-row vertical THT cut to 32 pins [7]), pre-routed to chain endpoints and NTC sense lines while leaving 24 pins user-jumperable; (c) the 6×6 DoE bond-pad array at 3.5 mm pitch with a co-located EIS calibration block (OPEN/SHORT/LOAD references with the $100\ \Omega \pm 0.1\%$ Vishay TNPW0603100RBEEA reference resistor [8]); (d) three TLM ladders ($W \in \{0.25, 0.5, 1.0\}$ mm, gaps $\{200, 300, 500, 1000, 2000, 4000\}$ μm) and four VDP cloverleaves ($W \in \{1.0, 0.5, 0.25, 0.15\}$ mm); (e) the LED row D1–D8 with the four Murata NCP15XH103J03RC NTC thermistors [9] interleaved between LED pairs, two daisy chains (DCL6 with 6 LEDs, DCL12 with 12 LEDs, both wired through the K_R cathode), and the 32-pin south header that pre-routes every D1–D8 signal (LED_VCC + 24 cathodes).

Design rule budget: 0.20 mm minimum copper track and 0.30 mm minimum copper clearance (both above the Aisler standard-pool 0.15 mm spec by $\geq 1.3\times$ and $2\times$ respectively); 0.30 mm minimum drill; 0.15 mm via annular ring. DRC: 0 violations, 0 unconnected items. ENIG plating (Ni 4 μm , Au 0.075 μm) is mandatory because every probe pad must be gold

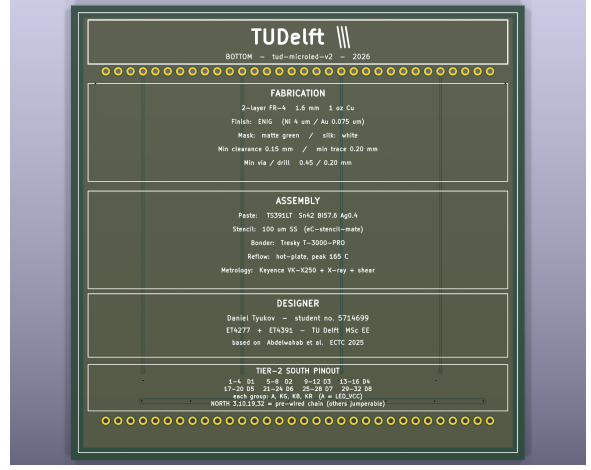
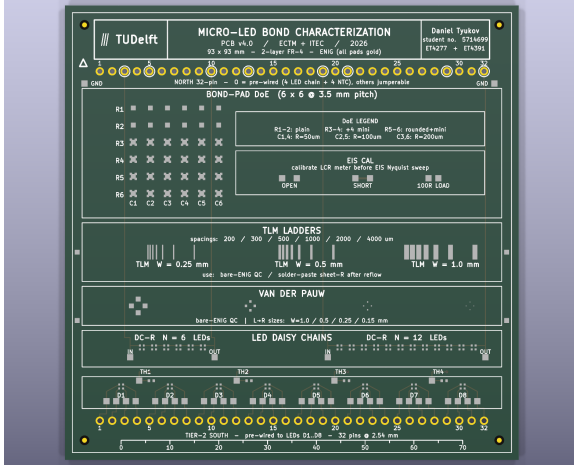


Fig. 1. v4.0.7 PCB, 93 mm $\times$ 93 mm, 2-layer FR-4, ENIG finish. Left: top side with the four functional zones (title block, DoE bond-pad array + EIS calibration, TLM/VDP test structures, LED row + NTCs) and 32-pin north/south breadboard headers. Right: bottom side with TUDelft branding, fabrication notes, designer block, and the south-header pinout legend that maps the 32 cable wires to the eight LED slots.

for reliable mechanical contact and to receive the bonded die. Total: 192 footprints — 69 fab-assembled (4 NTC, 1 resistor, 64 header pins) and 123 customer-handled (26 LED footprints bonded in cleanroom + 97 bare-ENIG test lands).

III. SCHEMATIC AND NET PLAN

The single-sheet A2 schematic in Fig. 2 captures the 36 electrically active components in six functional blocks and 164 named nets. Bare-pad test structures (TLM, VDP, DoE, probe, TC pads, fiducials, 123 entries) are documented in the design notes block and on the PCB silk rather than drawn out, since each has no connection beyond its own pad.

Net naming convention (top-level, for cross-referencing the BOM and PCB): LED_VCC is the 8-anode common bus (B.Cu copper pour); LED n _K{R,G,B} are the individual D n cathode signals; DCL{6,12}_IN/OUT are chain endpoints; DCL{6,12}_Jk are internal junctions; DCL n _Lk_K{G,B} are the per-chain-LED isolated green/blue cathodes; NTC n are the four thermistor sense lines; EIS_{LOAD, OPEN, SHORT}_{A,B} are the calibration probe pads; GND is the common return on the B.Cu pour.

IV. ELECTRICAL CHARACTERIZATION METHODS

Four orthogonal methods exercise the board:

A. Forward-voltage thermometry (V_F -TSP)

A diode's forward voltage at a fixed low current decreases monotonically with junction temperature; the slope (Temperature Sensing Parameter) $S \equiv -\partial V_F / \partial T_J$ is approximately constant for any given device and bias current and is the basis of the standard JEDEC JESD51-1 junction-temperature measurement [10], [11]. For an LED biased at a sense current $I_S \ll I_{op}$,

$$T_J(t) = T_{J,0} - \frac{1}{S} [V_F(t) - V_{F,0}], \quad (1)$$

where $V_{F,0}$ is the forward voltage at the calibration temperature $T_{J,0}$.

The four NTCs (TH1–TH4, Murata NCP15XH103J03RC, $R_{25} = 10 \text{ k}\Omega$, $B_{25/85} = 3380 \text{ K}$) sit between the LED pairs and provide the substrate-side reference; we read the LED V_F across the four bonded contacts via the south header and the NTC resistance via the north header. The NTC resistance maps to temperature through the Steinhart-B equation,

$$\frac{1}{T} = \frac{1}{T_0} + \frac{1}{B} \ln\left(\frac{R}{R_0}\right), \quad (2)$$

with $T_0 = 298.15 \text{ K}$, $R_0 = 10 \text{ k}\Omega$, $B = 3380 \text{ K}$ [9]. The pulsed sensing protocol of Pangallo et al. [12] avoids self-heating bias; we use $I_S = 100 \mu\text{A}$ pulses of 1 ms with 10% duty.

The bond-joint thermal resistance is extracted from the transient $Z_{th}(t)$ response after a step in operating current, deconvolved per Schwan et al. [13] or via the network-identification method of Liu et al. [14]. The relevant figure of merit is the steady-state bond resistance

$$R_{th,bond} = \lim_{t \rightarrow \infty} \frac{T_J(t) - T_{NTC}(t)}{P_{op}}, \quad (3)$$

with P_{op} the dissipated electrical power.

B. Electrochemical impedance spectroscopy (EIS)

EIS sweeps a small AC voltage (typically 10 mV) over 10^{-1} Hz to 10^6 Hz and records the complex impedance $Z(j\omega) = Z' + jZ''$. The Nyquist plot $Z''(Z')$ separates the bond-joint resistance (high-frequency intercept, $Z' \rightarrow R_s$ as $\omega \rightarrow \infty$) from the diode's parallel RC dynamics (low-frequency semicircle). For an ideal LED equivalent circuit (series bond resistance R_s , parallel diffusion resistance R_p with junction capacitance C_j),

$$Z(j\omega) = R_s + \frac{R_p}{1 + j\omega R_p C_j}. \quad (4)$$

Meaningful EIS requires instrument calibration. The board carries three on-PCB references that the user shorts the

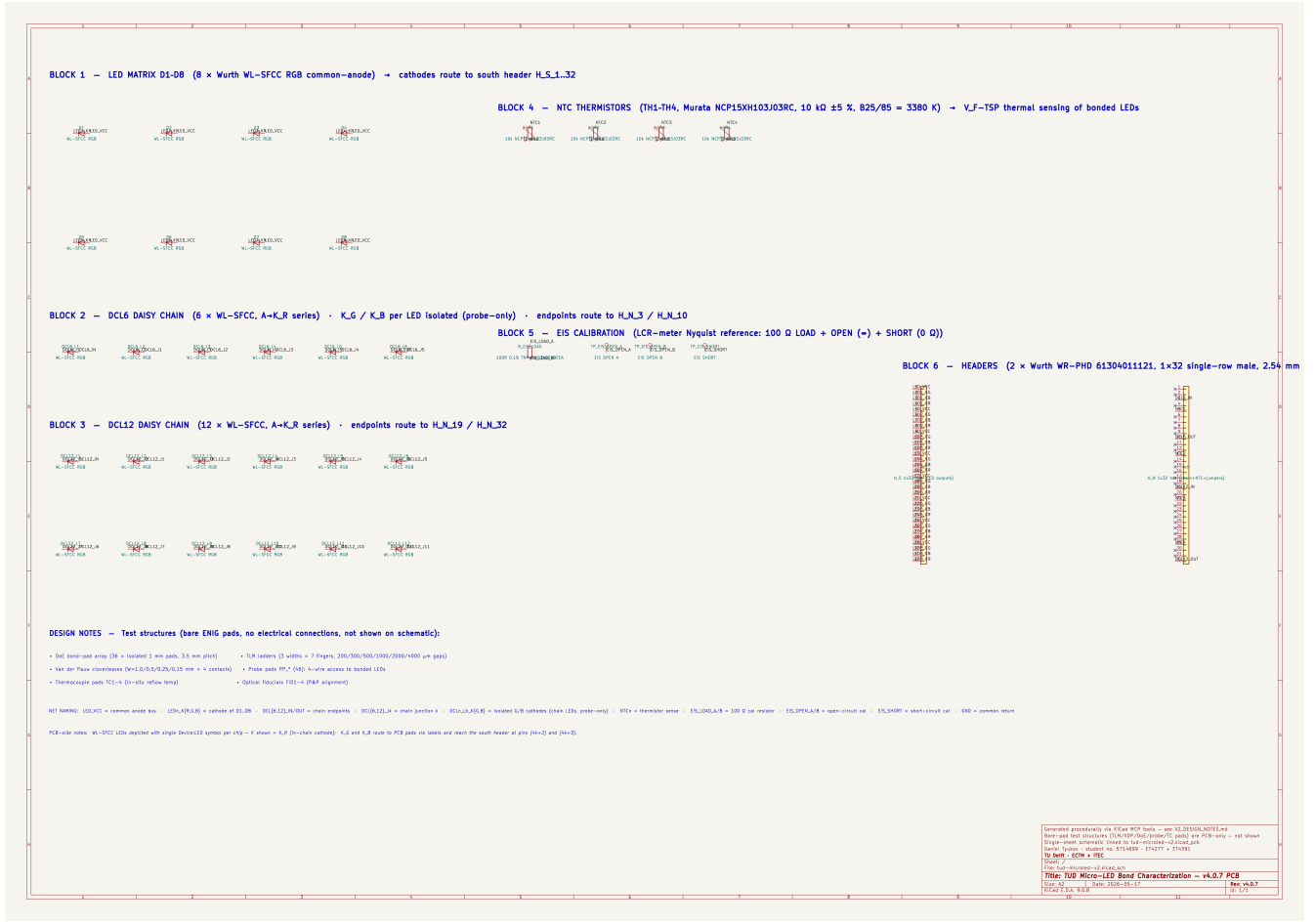


Fig. 2. Single-sheet A2 schematic. Block 1: D1–D8 common-anode LED matrix (cathodes routed to south header). Blocks 2–3: DCL6 and DCL12 daisy chains (A→K_R series; K_G/K_B per chain LED kept isolated for probe-only access). Block 4: TH1–TH4 NTC thermistors for V_F-TSP thermometry. Block 5: EIS calibration block (LOAD/OPEN/SHORT references). Block 6: north + south 32-pin headers with all routed pins labeled and no-connect markers on the 24 user-jumperable north pins.

LCR/EIS analyzer to before each sweep: EIS_OPEN (two isolated 1.27-mm gold pads, $Z \rightarrow \infty$), EIS_SHORT (paired pads joined by a solder bridge, $Z \rightarrow 0$), and EIS_LOAD (the 100 Ω ±0.1% Vishay TNPW0603100RBEEA thin-film resistor with 25 ppm/K TCR [8]). The 1.27-mm square pads are oversized relative to IPC-7351 0603 nominal so they double as four-wire Kelvin probe lands.

C. Transmission Line Model (TLM)

TLM extracts the sheet resistance R_{sh} of the metallization and the specific contact resistivity ρ_c between the bond pad and the LED electrode from the resistance measured between successive finger pairs at known gaps d [3], [4]. For N identical contacts of width W and characteristic length L_T ,

$$R_{tot}(d) = \frac{R_{sh}}{W} d + 2R_c, \quad R_c = \frac{R_{sh} L_T}{W} \coth\left(\frac{L}{L_T}\right), \quad (5)$$

where R_c is the contact resistance and L the contact length. A linear fit of R_{tot} vs d yields R_{sh} from the slope and $2R_c$ from the intercept; ρ_c then follows from $\rho_c = R_c \cdot W \cdot L_T \cdot \tanh(L/L_T)$.

Our three ladders span $W \in \{0.25, 0.5, 1.0\}$ mm (so the contact-width dependence of the transfer length can also be tested) and gaps $\{200, 300, 500, 1000, 2000, 4000\}$ μm. The smallest gap (200 μm) is above the Aisler standard-pool fabrication minimum, so all ladders are manufacturable in the same run as the LED-side traces and the measurement references the same ENIG surface that the bond contacts see.

D. Van der Pauw (VDP) sheet resistance

VDP four-point measurement on a symmetric cloverleaf gives a direct absolute R_{sh} value, independent of contact geometry, from the implicit relation

$$\exp\left(-\pi \frac{R_A}{R_{sh}}\right) + \exp\left(-\pi \frac{R_B}{R_{sh}}\right) = 1, \quad (6)$$

with R_A and R_B the two transverse four-point resistances measured at 90° rotation around the sample [4]. For a symmetric clover ($R_A \approx R_B \equiv R$), $R_{sh} = (\pi/\ln 2) R$. Our four cloverleaves ($W \in \{1.0, 0.5, 0.25, 0.15\}$ mm) cross-check the TLM slope and quantify width-dependent edge effects.

E. Daisy-chain bond-yield QC

The DCL6 and DCL12 chains place 6 (resp. 12) WL-SFCC LEDs in series via their K_R cathode, exercising $2N$ bonds per chain (anode + K_R cathode on each die). A current-voltage sweep across the chain endpoints (DCL n _IN/OUT on the north header) detects any open or high-resistance bond as a discontinuity in $I(V)$ or as a chain V_F that significantly exceeds the expected $N \cdot V_{F,nominal}$. For WL-SFCC red at 10 mA, $V_{F,R} \approx 2.05$ V [15], so DCL12 yields a clean go/no-go criterion around 24–25 V drive at 10 mA. The K_G and K_B cathodes of each chain LED are routed to isolated probe pads so the chain pass/fail can be cross-checked per-die and per-color.

V. TOOLING AND ASSEMBLY WORKFLOW

Software. KiCad 9.0.8 for the project (tud-microled-v2.kicad_pro); the .kicad_pcb is procedurally generated by tools/generate_pcb_text.py and the schematic by the same KiCad MCP tool chain that built Figure 2. Fab outputs are exported with kicad-cli: gerbers + drill (Aisler reads native .kicad_pcb but the bundle is shipped as a fallback), pick-and-place CSV, top/bottom review PDFs, the 3D STEP model, and the Beagle-format BOM (tools/gen_aisler_bom.py).

Fabrication. The board ships to Aisler (Aachen/Eindhoven) for the standard-pool 2-layer FR-4 1.55 mm process with ENIG. Beagle assembly soldering covers the 4 NTCs, the 1 EIS load resistor, and the 64 pin-header pins (2 strips of 32, cut from 1×40). The 26 WL-SFCC LED footprints are flagged *do not populate* (DNP) in the BOM and assembly notes, because they are the research subject and are bonded by the customer in the TU Delft cleanroom under controlled paste / reflow-profile / die-bonder conditions. Expected cost: ~115–125 for three fully assembled boards delivered.

Bonding. LED placement uses the Tresky T-3000-PRO die bonder (envelope 95 mm × 95 mm, our 93 mm board fits with 1 mm margin). Solder paste is stencilled through a 100 μ m laser-cut stencil keyed to the WL-SFCC pad layout (4 corner pads, 0.4 mm × 0.4 mm at 0.8 mm pitch per the Würth recommendation [15]). Reflow follows IPC J-STD-020E with peak temperature 245–260 °C per the solder paste vendor.

Measurement. Lab equipment used in our cleanroom: Keithley 2450 SourceMeter for V_F -TSP and I-V sweeps; Solartron 1260A FRA + 1296A dielectric interface for EIS; HP4156 parameter analyzer for TLM/VDP; Tresky as bonder; T-type thermocouples soldered to the four TC_n gold pads for in-situ reflow temperature logging. Probe access is via 1.27 mm Kelvin probes on the PP_* pads or via ribbon-cable breakouts from the 32-pin headers.

VI. SOLDER-MIXTURE EXPERIMENT PLAN

The board’s primary research target is to compare commercial solder pastes of different chemistries on identical LED-and-substrate hardware. We nominate four candidates that span the relevant trade space:

- 1) **SAC305** (Sn 96.5% / Ag 3.0% / Cu 0.5%) — industry baseline.
- 2) **SAC0307** (low-silver variant) — lower cost, slightly worse thermal cycling, similar V_F drop.
- 3) **Au-Sn 80/20 eutectic** — high-temperature reference [3] with documented best-in-class contact resistivity.
- 4) **Polymer-reinforced SAC305** — emerging impact-resistant chemistry [1] relevant to flexible-display applications.

For each paste we run a 3-board batch (one trial + one real + one spare, matching Aisler’s bundling). On each board we measure: chain-yield (N_{ok}/N across DCL6 and DCL12), per-LED V_F at 10 mA, $R_{th,bond}$ from V_F -TSP transient, $R_s + R_p + C_j$ from EIS fit to (4), and ρ_c from TLM fit to (5). Cross-board variance gives statistical bounds on each paste’s process window.

Beyond the solder-mixture comparison the board also supports investigations that re-use the same characterization plumbing: capillary self-alignment yield as a function of pad geometry (the DoE 6×6 array with its plain / minis / rounded variants), reflow profile optimization (T_C vs T_{C-5} tracking on the TC pads), and bond ageing studies (the chains can be loaded with a programmable current source to drive thermomechanical fatigue).

VII. CONCLUSION

We have presented a self-contained 93 mm × 93 mm PCB test vehicle that consolidates four electrical-characterization methods (V_F -TSP thermometry, EIS, TLM, VDP) plus daisy-chain yield QC on a single substrate, enabling a complete bond-quality fingerprint from one cleanroom session. The board is fab-ready on the Aisler standard pool with ENIG plating, with a 0-DRC-violation budget and conservative 2× margin over the published manufacturing minima. The accompanying schematic, BOM, and tooling pipeline make the design reproducible. Part 2 of this report will present the first electrical-characterization results from the four-paste comparison once the boards arrive and the cleanroom bonding session is complete.

ACKNOWLEDGMENT

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